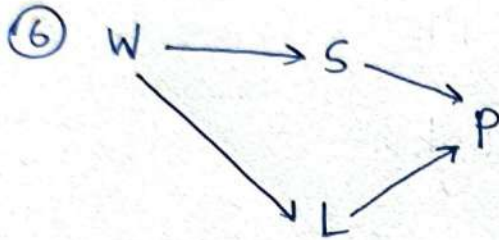


Healthy score ≈ 0.0728

Diseased score = 0

Since, $0.0728 > 0$

The classifier predicts 'Healthy'.



A software company uses Bayesian Network to predict whether an employee will complete a project on time based on workplace conditions.

The network contains 4 binary variables:

W = Heavy Workload (Yes/No)

S = High Stress (Yes/No)

L = Works Late Hours (Yes/No)

P = Project completed on Time (Yes/No)

Heavy Workload (W) Probability

Yes	0.4
No	0.6

W	$P(S=\text{Yes} W)$
Yes	0.8
No	0.3

W	$P(L=\text{Yes} W)$
Yes	0.7
No	0.2

S	L	$P(P=\text{Yes} S, L)$
Yes	Yes	0.4
Yes	No	0.6
No	Yes	0.75
No	No	0.95

① Calculate the joint probability of the following event:

$$P(W = \text{Yes}, S = \text{Yes}, L = \text{Yes}, P = \text{Yes})$$

Answer:

Given Bayesian Network,

$$W \rightarrow S, \quad W \rightarrow L$$

$$S, L \rightarrow P$$

The joint probability is:

$$P(W, S, L, P) = P(W) P(S|W) P(L|W) P(P|S, L)$$

Using the given probabilities:

$$P(W) = 0.4$$

$$P(S|W) = 0.8$$

$$P(L|W) = 0.7$$

$$P(P|S, L) = 0.4$$

$$\therefore P(W, S, L, P) = 0.4 \times 0.8 \times 0.7 \times 0.4$$
$$= \boxed{0.0896}$$

② Calculate the probability that an employee completes the project on time.

Answer:

Since W, S, L are not specified, we sum over all possible combinations:

$$P(P=Y) = \sum_{W, S, L} P(W) P(S|W) P(L|W) P(P|S, L)$$

$$\begin{aligned}
&= 0.4 \times 0.8 \times 0.7 \times 0.4 \\
&+ 0.4 \times 0.8 \times 0.3 \times 0.6 \\
&+ 0.4 \times 0.2 \times 0.7 \times 0.75 \\
&+ 0.4 \times 0.2 \times 0.3 \times 0.95 \\
&+ 0.6 \times 0.3 \times 0.2 \times 0.4 \\
&+ 0.6 \times 0.3 \times 0.8 \times 0.6 \\
&+ 0.6 \times 0.7 \times 0.2 \times 0.75 \\
&+ 0.6 \times 0.7 \times 0.8 \times 0.95
\end{aligned}$$

$$\begin{aligned}
&= 0.0896 + 0.0576 + 0.042 + \\
&0.0228 + 0.0144 + 0.0864 + \\
&0.063 + 0.3192
\end{aligned}$$

$$= 0.695$$

$$\therefore P(P=\text{Yes}) = \boxed{0.695}$$